



European **Hyperloop Center**

A new testing ground bolstering the collaboration
of the hyperloop industry



User Perceptions of Hyperloop

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Interpreting Previous Research

	Purpose	Method	Advantages	Disadvantages
Shetty (2019)	What are the different public perspectives on the development of Hyperloop as a mode of transport?	Designed a Q-methodology to measure subjective opinions about the hyperloop	Gathered unique perspectives about the hyperloop	Small sample size, low reliability and validity with Q-methodology
Planning et al. (2020)	What is the level of knowledge and acceptance regarding the hyperloop?	Large survey analysis	Large and representative sample size of the NL population	Lacks first hand experience with hyperloop
Altena (2021)	How do people experience the spaciousness of the proposed hyperloop S and which properties influence the experience of the user?	Measured initial reactions to mock-up and comparisons to small and large model using VR headset.	VR Hyperloop experiment, participants could compare multiple mock-up dimensions	Unrepresentative sample of the general population



Research Objectives + Methodology

Our objective is to measure user perceptions in a simulated hyperloop experience:

- How do users perceive the current dimensions of the hyperloop?
- What elements of the hyperloop do users see as comfortable/uncomfortable?
- Do users perceive the hyperloop as a safe mode of transportation?
- How likely are users to travel by hyperloop once it is available? (repatronage)

Materials

- Mock-up Hyperloop
- VR Headset
- Online questionnaire (Google forms)



Sampling Method

- Advertisement via email and Social Media
- Participants signed up by completing consent form
- Once given consent, they were emailed a set of available testing dates to attend.

**Sign up to participate in
Hyperloop Research in
Groningen!**



Together with students of the Rijksuniversiteit Groningen we start with consumer research to gain insight on the public's perceptions while travelling in the hyperloop.

You will have the opportunity to sit inside a mock-up capsule while using a VR headset and experience a simulation travelling by hyperloop. You will be asked to answer a questionnaire regarding your perceptions and sensations during the simulation.

The study should take approximately 15 minutes, on the location of Bytesnet (Zernike Campus).

Sign up here!



<https://tinyurl.com/2y7z5ycv>



Procedure



- Participant is met outside the building by the experiment proctor and guided to the experiment.
- The proctor explains the overview of the study and participant is first asked to complete the Part 1 questionnaire.
 - (Demographics, Tech and Hyp Attitudes)
- After baseline questions are answered, the participant puts on the VR headset and enters the mock-up hyperloop.
- Proctor allows participant to spend time in mock-up with VR headset on.
- Participant answers remaining questions regarding their sensory experience in the hyperloop and their opinions of safety and repatronage.



Measures

Five separate scales were used by the researchers

Technology Attitudes (TA)

- Milfont & Duckitt, (2010). Ten item Likert Scale 05: Confidence in Science and Technology.
- Measures a user's confidence in technology solving complex environmental issues.

Hyperloop Attitudes (HA)

- Shetty, (2019). Eight item Likert-Scale: Adoption of Hyperloop.
- Measures a user's values and perceptions of the hyperloop.

Safety

- Self-constructed. Four item Likert-scale.
- Measures the user's perceptions of security and worry.

Repatronage

- Self-constructed. Three item Likert-scale.
- Measures a user's intention to travel by hyperloop in the future.

Comfort

- Self-constructed. Multiple factors: one item Likert-scale per factor.
- Measures how pleasant/unpleasant the simulated hyperloop experience was by assessing multiple factors.

Participants were also given the opportunity to provide feedback regarding their hyperloop experience via short answer question.

Factors: claustrophobia, comfort for a 1-hour travel time, comfort for a 2-hour travel time, preferred height, preferred width, leg space, eating and computer usage.



Method of Analysis

Data was used as an exploratory analysis.

Researchers performed descriptive statistics on the five measures. In this presentation, we will discuss:

- The gender and age distributions in the sample
- Normality and trends with the pre-test data of TA and HA
- Comfort item means and standard deviations
- Correlations between age, gender, TA, HA, safety, and repatronage.
- Frequencies of positive and negative observations in the free response question



Method of Analysis

Scoring

Participants were asked to rate how much they agree with each statement on a scale from 1 - 5.

- A participant's total score for TA, HA, Safety, and Repatronage was a computed average of the item scores
 - For example: $(\text{total SUM of TA scores}) / (\text{number of items}) = \text{TA score}$

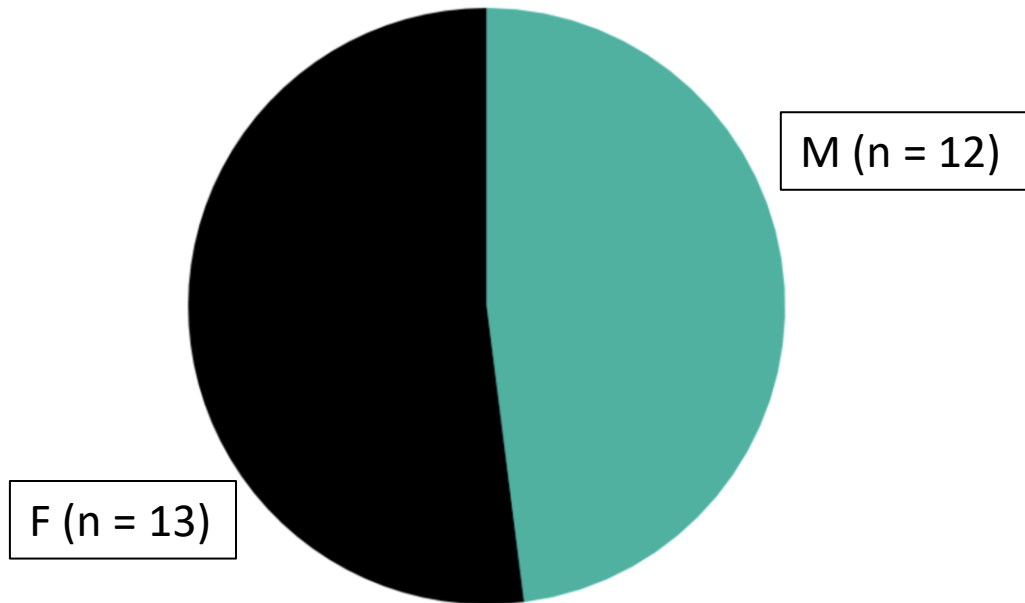
De meeste milieuproblemen kunnen worden opgelost door meer en betere technologie toe te passen./Most environmental problems can be solved by applying more and better technology.

	1	2	3	4	5	
Zeer oneens/Strongly Disagree	☐	☐	☐	☐	☐	Sterk mee eens/Strongly Agree



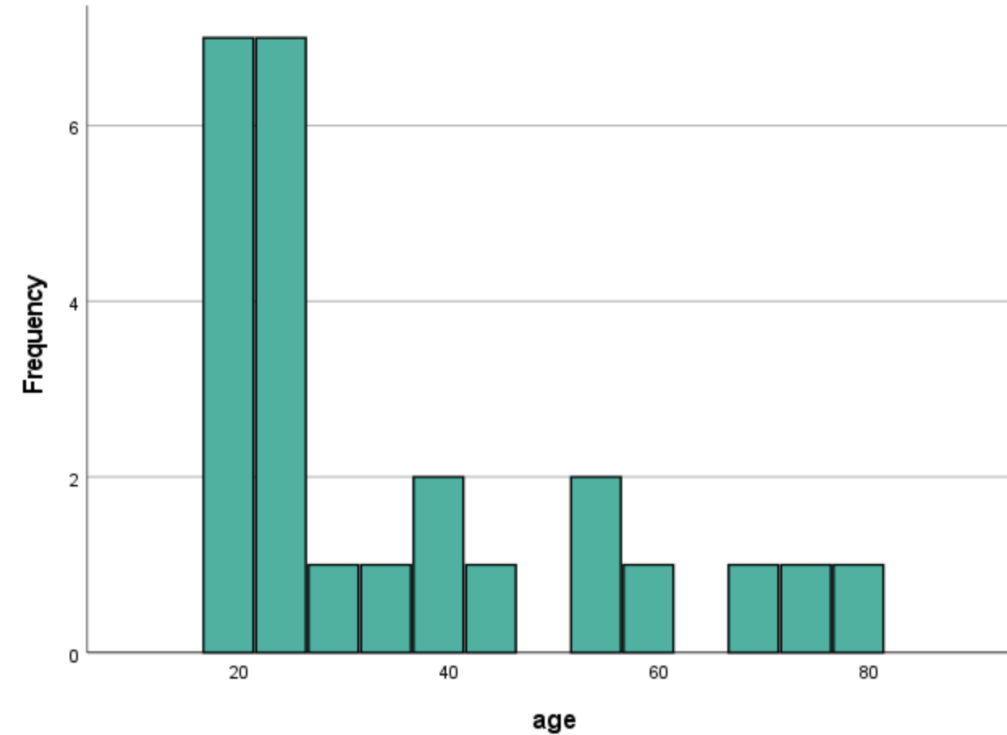
Demographic Information

Gender Distribution



- Gender distribution close to equal

Age Distribution

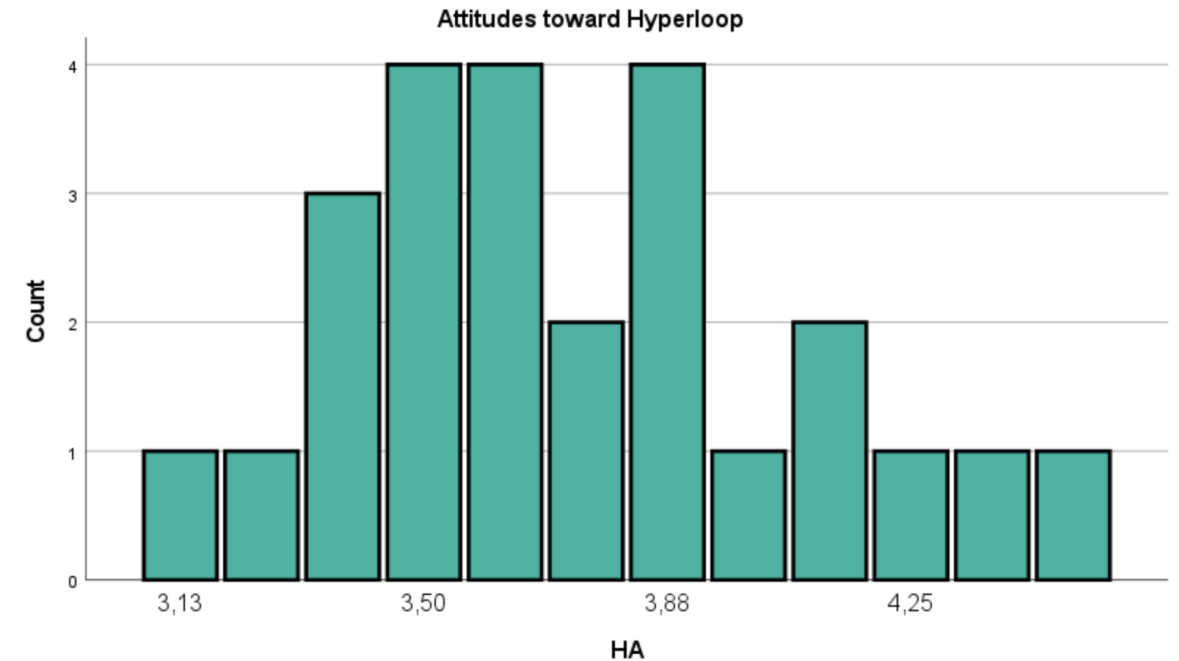
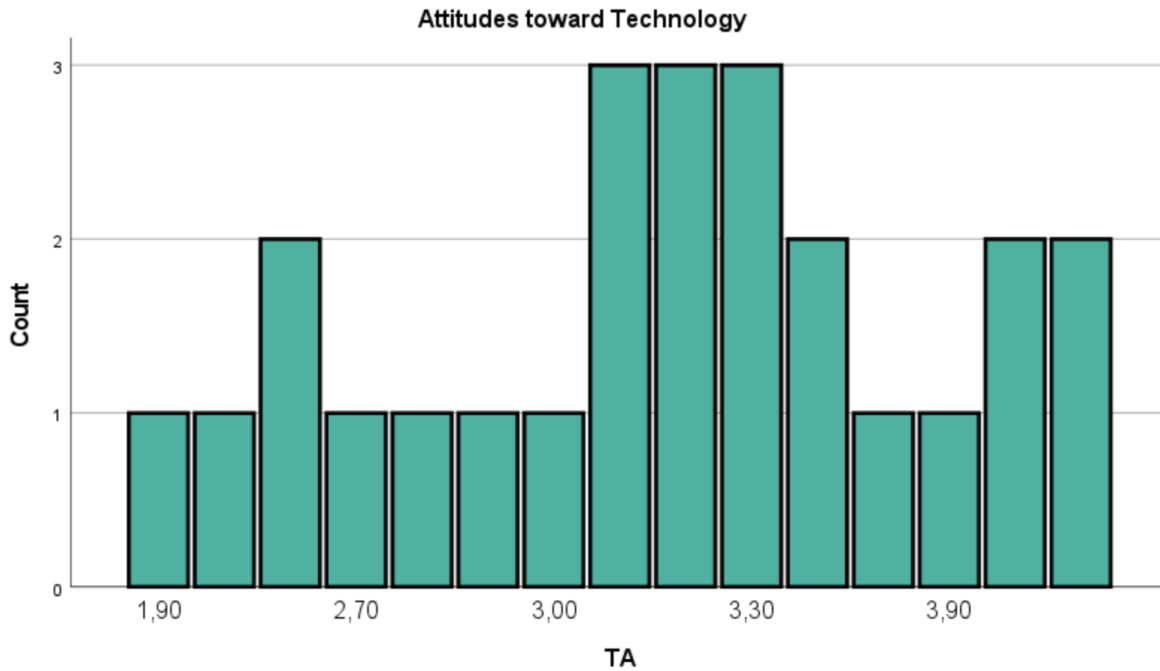


- Mean: 35
- Standard Deviation: 19
- Mode: 20 and 21
- Min: 20 / Max: 79



Attitudes

Personal opinions regarding technology and the environment were collected before hyperloop simulation.



Descriptive Statistics

	N	Range	Minimum	Maximum	Mean	Std. Deviation	Variance	Skewness	Std. Error
Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic
TA	25	2,20	1,90	4,10	3,1720	,64841	,420	-,479	,464
HA	25	1,50	3,13	4,63	3,7450	,37410	,140	,691	,464
Valid N (listwise)	25								

- Scores range from 1 -5
- 1 = attitudes not in favor
- 5 = attitudes in favor

- On average, participants were more favorable towards both technology and hyperloop before completing the experiment.

- People's views on technology varied more than with the hyperloop.
- People were generally more in favor of the hyperloop than technology.



Interpreting Means and Standard Deviations

- Little to no claustrophobia was experienced.
- Participants showed average anticipated comfort for 1-hour and 2-hour travel times.
 - Large SD may be concerning
- Participants preferred the dimensions to be slightly higher and wider.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Claustrophobia	25	1	4	1,40	,707
1 hour ride	25	1	5	3,52	1,358
2 hour ride	25	1	5	3,12	1,269
height	25	2	5	3,16	,554
width	25	2	4	3,28	,542
leg space	25	1	5	2,84	1,405
eat and computer usage	25	2	5	3,56	1,121
Valid N (listwise)	25				

- Perceptions of leg space were negative, and the distribution was highly variated
 - The large variation could be explained by differences of height in the sample.
- Most participants had positive perceptions of eating and computer usage space.



Interpreting Correlations

Criterion Variables: Safety and Repatronage

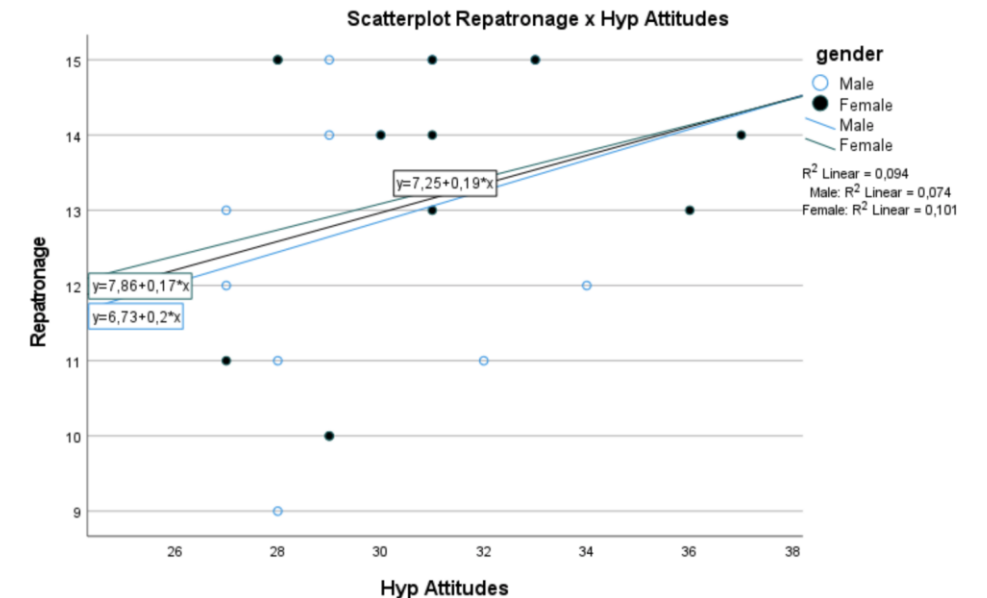
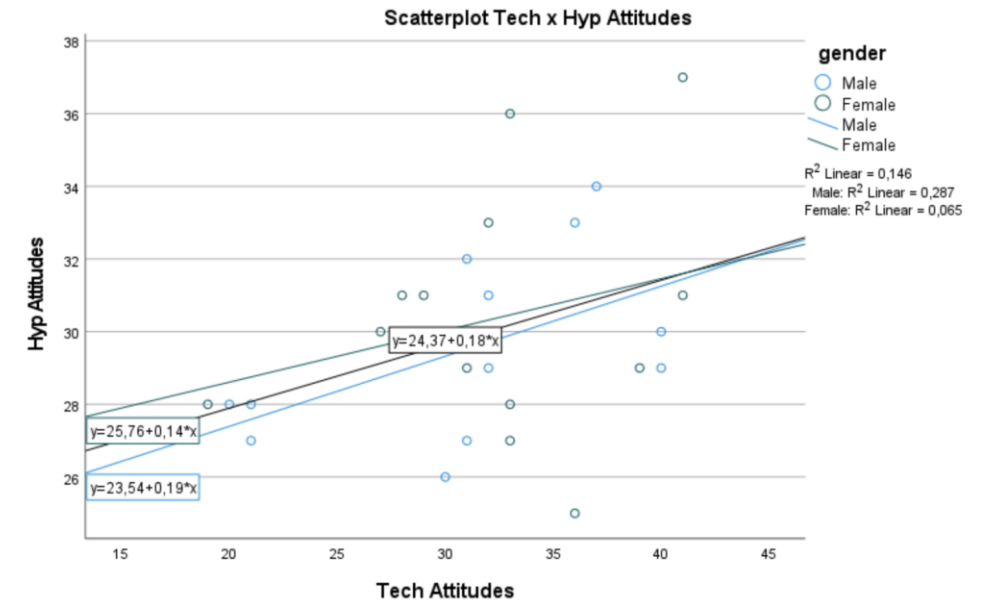
- We noticed moderate correlations with TA-HA, HA-Repatronage, and Safety-Repatronage.

Confidence Intervals

	Pearson Correlation	Sig. (2-tailed)	95% Confidence Intervals (2-tailed) ^a	
			Lower	Upper
Tech Attitudes - Hyp Attitudes	,382	,060	-,016	,675
Tech Attitudes - Safety	,162	,439	-,249	,524
Tech Attitudes - Repatronage	,172	,411	-,240	,531
Hyp Attitudes - Safety	,035	,869	-,365	,424
Hyp Attitudes - Repatronage	,307	,136	-,101	,626
Safety - Repatronage	,586	,002	,248	,797

a. Estimation is based on Fisher's r-to-z transformation.

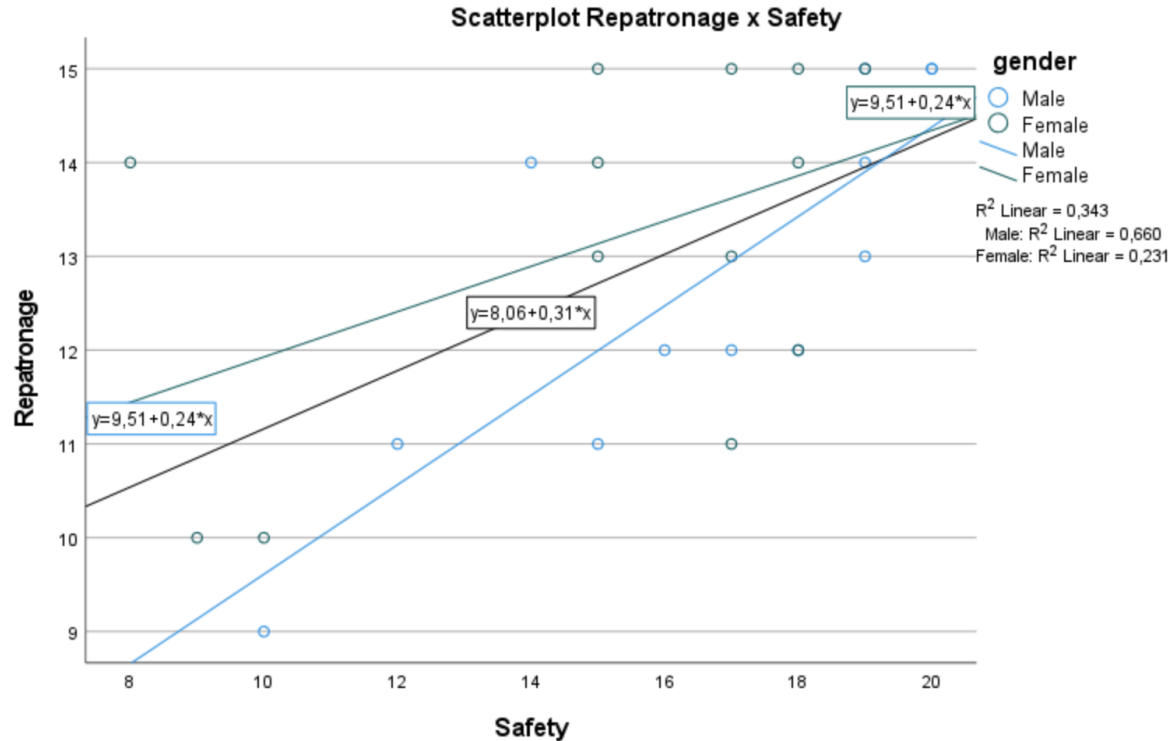
- Pearson Correlation (R^2): measures strength of a linear relationship between two variables; 0 means no correlation, values closer to 1 or -1 means there is a relationship.





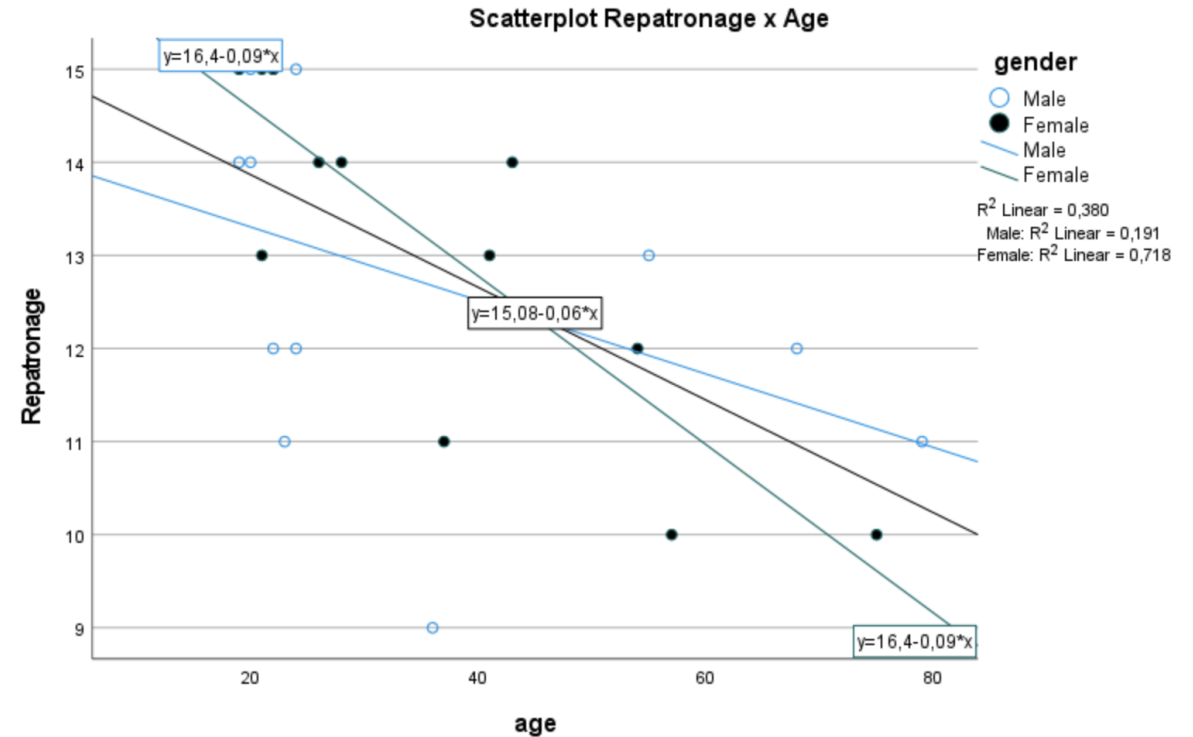
Interpreting Correlations

Criterion Variables: Safety and Repatronage



- Users who were more confident in the safety of the transport were more likely to travel by hyperloop once it is available.

- Age was the highest predictor of repatronage.
- Older participants were less likely to travel by hyperloop once it becomes available.



[illegible]

skyline	8
tables	7
leg space	7
charging port	5
luggage storage	3
comfortable seats	2
wider	2
larger walking path	2
travel map	1
toilets	1
headrest	1
seatbelts	1
chairs face in same direction	1
higher	1
arm rest	1



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Discussion: What was good? Bad?

Sample

Good:

- Mostly equal distribution of gender and age
- Predisposed attitudes had small effect on results

Bad:

- Small sample size
- Recruiting participants was difficult during testing period
- Advertisements for recruiting participants were seen by a selective audience already interested in hyperloop

Statistical Methods

Good:

- Descriptive statistics revealed possible connections between attitudes, safety, and repatronage; which can prompt future research interests

Bad:

- Cannot infer causality
- Only 2/5 scales were previously used by other researchers

Experiment

Good:

- Combination of the mock-up and VR headset gave participants a realistic view of the interior of the hyperloop

Bad:

- Headset did not include potentially important characteristics such as vehicle movement or perception of other passengers



Discussion: Conclusions

- How do users perceive the current dimensions of the hyperloop?
 - Participants were generally pleased with the current dimensions.

- Do users perceive the hyperloop as a safe mode of transportation?
 - Yes, but these results should be treated with caution because of the positive correlation with HA.

- What elements of the hyperloop do users see as comfortable/uncomfortable?
 - Leg space was the most valued in the quantitative analysis.
 - Free response item also revealed that artificial light (skyline) and tables were desired for comfort.

- How likely are users to travel by hyperloop once it is available? (repatronage)
 - Likelihood was above average (Item scores 1 - 5. Mean = 4.32, SD = 0.61)



Discussion: Conclusions

- Due to our small sample size and correlational data, it is unlikely we can infer too much from the results.
- Older populations may be a population of interest in future research.
- The results indicate there are considerable trends between HA, safety, age, and the criterion variable repatronage.



Discussion: Future Research?

- Increase sample size.

- Develop more vivid and realistic hyperloop virtual realities.
 - (e.g., create perception of movement, other passengers, and interior characteristics)

- Recruit sample from multiple different sources to avoid sampling bias.

- Design experimental methods that can infer causality in the research.



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